7.6.3.1 Magnification in astronomical telescope

The magnification m is the ratio of the angle β subtended by the image to the angle α subtended by the object with the principal axis.

$$m = \frac{\beta}{\alpha} \qquad (7.80)$$

From the diagram, $\alpha = \frac{h}{f_0}$ and $\beta = \frac{h}{f_e}$

$$m = \frac{f_o}{f_e} \qquad (7.81)$$

The length of the telescope is approximately,

$$L = f_o + f_e \qquad (7.82)$$

EXAMPLE 7.21

A small telescope has an objective lens of focal length 125 cm and an eyepiece of focal length 2 cm. (a) What is the magnification of the telescope? (b) What is the separation between the objective and the eyepiece? (c) What is the angular separation between two stars when viewed through this telescope if they subtend 1′ for bare eye?

Solution

f_o = 125 cm; f_e = 2 cm; m = ?; L = ?; θ_i = ?

(a) Equation for magnification of telescope,

$$m = \frac{f_o}{f_e}$$

Substituting, $m = \frac{125}{2} = 62.5$

(b) Equation for approximate length of telescope, $L = f_o + f_e$

Substituting, L = 125+2 = 127 cm = 1.27 m

(c) Equation for angular magnification,

$$m = \frac{\theta_i}{\theta_0}$$

Rewriting, $\theta_i = m \times \theta_0$

Substituting,

$$\theta_i = 62.5 \times 1' = 62.5' = \frac{62.5}{60} = 1.04^\circ = 1°2'30''$$

7.6.4 Terrestrial telescope

A terrestrial telescope is used to see objects at a long distance on the surface of earth. Hence, image should be erect. Hence, it has an additional erecting lens to make the final image erect as shown in Figure 7.41.

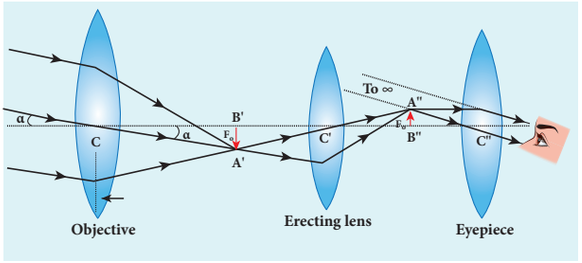


Figure 7.41 Terrestrial telescope

7.6.5 Reflecting telescope

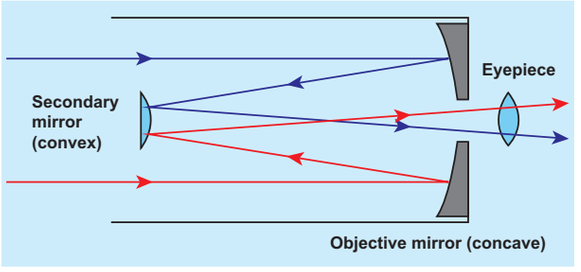


Figure 7.42 Reflecting telescope

Modern telescopes use concave mirrors instead of lenses for the objectives. It is rather difficult and expensive to make lenses of large size which form images that are free from any optical defect. **A telescopes which has a concave mirror objective is called *reflecting* telescope.** It has several advantages. Only

XII_Physics_Vol-2_UNIT-7_EM.indd 92 03-03-2022 15:19:34